

Original article

Modeling heterogeneity in air pollution mixture effects on birth weight: A spatially varying coefficient approach

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ARTICLE INFO

Keywords:

Air pollution

Birth weight

Environmental exposure

Epidemiologic Methods

Maternal Exposure

Spatial Analysis

ABSTRACT

Purpose: To extend the existing quantile g-computation framework for studying environmental exposure mixtures to estimate local effects of ambient air pollution mixtures on birth weight. This framework has traditionally been applied to estimate global mixture effects without accounting for spatial heterogeneity.

Methods: First, pregnancy-wide maternal exposure to five common air pollutants is estimated for nearly 1.5 million births occurring in Georgia, USA between 2005 and 2016. Then, a recently developed spatially varying coefficient model based on Bayesian additive regression trees (BART) is applied to estimate spatially heterogeneous mixture effects using quantile g-computation. Results are compared with those obtained from traditional conditional autoregressive models, as well as spatially agnostic modeling approaches.

Results: We find evidence of county-level spatially varying mixture associations, where for 21 of 159 counties in Georgia, elevated concentrations of a mixture of PM_{2.5}, nitrogen dioxide, sulfur dioxide, ozone, and carbon monoxide were associated with a reduction in birthweight by as much as -14.77 grams (95% credible interval: -21.24, -9.78) per decile increase in all five air pollutants.

Conclusions: Spatially varying coefficient models based on BART outperform alternative approaches when modeling the relationships between air pollution mixtures and birth weight for the majority of counties in Georgia.

Introduction

In 2022, the National Center for Health Statistics reported that an estimated 8.60% of infants born in the United States had low birth weight (less than 2,500 grams) [1]. Low birth weight has a strong association with infant mortality and morbidity. In the same year, infant mortality rate in the United States was 42.36 per 1000 live births among low birth weight infants, compared to just 2.10 per 1000 live births among infants greater than 2500 grams [2]. Identifying risk factors of reduced birth weight, particularly those due to modifiable environmental risk factors, is an important research priority.

In environmental and perinatal epidemiology, there is a rich literature supporting the associations between various air pollutants and birth outcomes including, but not limited to, reduced birth weight [3–6]. Studies have frequently identified associations between low birth weight and elevated concentrations of PM_{2.5}, nitrogen dioxide, sulfur dioxide, ozone, and carbon monoxide, among others. We have previously reported these associations in Atlanta, Georgia [7,8].

In the past, studies of association between air pollution and health outcomes have utilized single-exposure models. While useful, these

models may be inadequate for describing the combined effect of multiple air pollutants that individuals are simultaneously exposed to. In recent years, research has shifted toward developing and applying mixture models that attempt to quantify this joint association between multiple exposures and health. Of the many modeling strategies introduced, quantile g-computation (QGCMP) proposed by Keil et al. [9] has been the most widely used approach in population-based epidemiologic studies. QGCMP is favored for its computational speed, simple definition and interpretation of the overall mixture effect, and its straightforward implementation via the well-maintained *qgcomp* R package [10].

In many studies of environmental mixtures where QGCMP or alternative approaches might be used, it is common to have health and exposure data acquired from a large geographical study region. While compiling data from all regions within the study area increases sample size and the ability to detect small mixture effects, it also provides an opportunity to explore spatial heterogeneity in health effects.

In this work, we consider a spatially varying coefficient (SVC) model based on Bayesian additive regression trees (BART) [11,12] to estimate

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Received 6 May 2025; Received in revised form 21 September 2025; Accepted 3 October 2025

Available online 10 October 2025

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heterogeneous mixtures effects within the QGCOMP framework. BART is a flexible modeling approach that has consistently performed well on a variety of prediction, classification, and causal inference tasks [11, 13, 14]. An additional benefit of using BART is that, unlike most other machine learning models, BART is fully Bayesian and thus offers natural uncertainty quantification via the posterior distribution. We compare the proposed approach with spatially varying coefficient models from disease mapping through an analysis of birth weight from vital records in the state of Georgia.

Data

Air pollution data

We considered five air pollutants: fine particulate matter with diameter 2.5 μm and smaller ($\text{PM}_{2.5}$, 24-h average, $\mu\text{m}/\text{m}^3$), nitrogen dioxide (NO_2 , 1-h max, ppb), sulfur dioxide (SO_2 , 1-h max, ppb), ozone (O_3 , 8-h max, ppm), and carbon monoxide (CO , 1-h max, ppb). Daily estimates of the concentrations of each pollutant were derived from a data fusion model which utilized simulations from the Community Multiscale Air Quality Model and monitoring data from the Environmental Protection Agency's Air Quality System database [15]. The original data product is available at a 12 km gridded spatial resolution. We used area-weighted averaging to obtain exposures at the ZIP code level.

Health data

We obtained birth records from the Office of Health Indicators for Planning, Georgia Department of Public Health. We restricted the data to only include singleton pregnancies with gestational age greater than 27 weeks and an estimated date of conception between January 1st, 2005 and December 31st, 2016. There were a total of 1,468,531 births meeting this criteria. Additional covariates collected on the birth mothers included age (years), race, level of educational attainment, marital status, and parity. Pregnancy-wide air pollution exposures were estimated by linking maternal residential address ZIP code and calculating the average concentration of each pollutant from the date of conception to the date of birth.

Methods

Review of quantile g-computation for mixture modeling

The goal of QGCOMP, like its predecessor weighted quantile sum, is to provide a more interpretable *mixture effect*. For this reason, QGCOMP is sometimes referred to as a *summary index* method. This stands in contrast to *response surface* methods, such as Bayesian kernel machine regression (BKMR), which provide a more flexible approach to modeling complex exposure-response surfaces, but generally are not as easily implemented or interpreted.

In the QGCOMP framework, the target parameter(s) quantify the expected change in the outcome from increasing all exposures of interest by one quantile. This provides a summary measure of the overall impact of a correlated exposure mixture, thereby avoiding the instability from multicollinearity that can distort the interpretation of individual exposure coefficients. Because the implementation of QGCOMP leverages model fitting procedures from standard regression models, it can be run efficiently and has been extended to a variety of models. This is particularly important for studies which make use of administrative datasets, for which computationally burdensome methods such as BKMR are impractical.

The linear and additive conditional model in the QGCOMP framework is given by (1):

$$Y_i = \beta_0 + \mathbf{x}_i^T \boldsymbol{\beta} + \mathbf{w}_i^T \boldsymbol{\gamma} + \epsilon_i \quad \epsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad (1)$$

where $\mathbf{x}_i$ is a P_x -vector of quantized exposures for observation i . Here, the term quantized exposure refers to an originally continuous exposure whose values have been recoded to $0, 1, \dots, Q-1$, where the new value represents which of the Q quantiles the original observed value belonged to. The model may also include $\mathbf{w}_i$, a P_w -vector containing confounders for adjustment for observation i (note these are not generally quantized). Thus, each element of $\boldsymbol{\beta} = (\beta_1, \dots, \beta_{P_x})^T$ represents the expected change in the outcome for a one quantile increase in the corresponding exposure, holding all other exposures and confounders fixed, while the elements of $\boldsymbol{\gamma}$ retain typical interpretations for regression coefficients.

The reasoning behind this treatment of the exposures of interest is to define the mixture effect as $\Psi = \sum_{p=1}^{P_x} \beta_p$. When the model is linear and additive in terms of the quantized exposures as in (1), Ψ represents the expected change in the outcome for a one quantile increase in every exposure. In this scenario, Ψ also coincides with the slope parameter from a simple linear marginal structural model (MSM) in which the sole predictor, denoted S_q , represents the joint quantized exposure level. Specifically, S_q takes on values $0, \dots, Q-1$, corresponding to when all quantized exposures are simultaneously set to $0, \dots, Q-1$. When framing the problem in this manner, Ψ may alternatively be estimated via g-computation with the joint exposure quantile S_q . In this work we focus on the linear and additive model (1), but in general estimation of Ψ via the MSM approach can be used when higher order terms are desired. Keil et al. [9] notes that when certain identifiability assumptions hold, Ψ might be interpreted as a causal parameter [9].

Spatially varying quantile g-computation with BART

We propose allowing the individual exposure coefficients $\boldsymbol{\beta}$ in model (1) to vary across space, which in turn implies the mixture effect Ψ also varies across space. The result is the conditional model in (2):

$$Y_i = \beta_0(z_i) + \sum_{p=1}^{P_x} \beta_p(z_i) \mathbf{x}_{i,p} + \mathbf{w}_i^T \boldsymbol{\gamma} + \epsilon_i \quad \epsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad (2)$$

where the intercept and quantized exposure coefficients depend on the spatial location of observation i , z_i . The local mixture effect specific to location z is then defined as $\Psi(z) = \sum_{p=1}^{P_x} \beta_p(z)$.

There are a few reasons allowing for a spatially varying air pollutant mixture effect is warranted. For example, the mixture of $\text{PM}_{2.5}$ components may differ spatially due to differences in local emission sources and meteorology. Also, spatially varying population characteristics that impact the relationships between personal exposure and ambient concentration may result in effect heterogeneity. When the target estimand is the overall mixture effect, differences may also be attributable to specific exposure levels due to a nonlinear exposure-response relationship. For example, NO_2 may be a more important component in the mixture for regions near highways where levels are high. Additionally, if the true exposure-response surface contains any interactions, then the effects of individual exposures and the overall mixture effect is likely different in regions with different exposure concentrations. Allowing for spatially varying weights allows for capturing locally linear mixture effects, even when the overall mixture effect is more complex.

We suggest placing BART priors on the spatially varying intercept and each of the exposure coefficients in model (2). While BART has traditionally been used to model an outcome, the varying coefficient BART (VCBART) model [12] extends the approach to model multiple regression coefficients directly. This approach allows each coefficient to vary flexibly over space, capturing complex patterns that may be difficult to represent with parametric alternatives. Formally, in VCBART we specify $\beta_p = \sum_{t=1}^T \text{Tree}(z; \mathcal{T}_t, \mathcal{M}_t)$, $p = 0, \dots, P_x$, where $\text{Tree}(z; \mathcal{T}, \mathcal{M})$ is the function which deterministically maps a vector of covariates z to a single leaf node μ of tree $\mathcal{T}$, and $\mathcal{M}$ represents the collection of leaf nodes. Then, if μ has prior $N(0, T^{-1}\sigma_\beta^2)$, the implied prior for $\beta_p(z)$

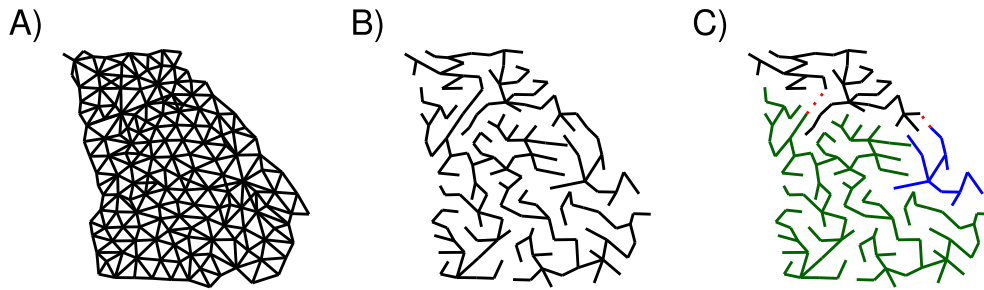


Fig. 1. Illustration of the SVC BART spatial branching process, applied to counties in Georgia. (A) Edges are drawn between centroids of adjacent counties. (B) A random spanning tree is sampled from the graph in panel A. (C) Three groups of contiguous counties are formed by deleting two edges from the spanning tree in panel B.

is $N(0, \sigma_\beta^2)$, and the for $\Psi(z)$ is $N(0, P_x \sigma_\beta^2)$. Further details regarding estimation of $\{\tau_i, \mathcal{M}_i\}_{i=1}^T$ are described in Deshpande et al. [12].

Like other tree-based methods, BART partitions the covariate space into regions via “splits” on continuous, ordinal, or categorical data. For spatial data, Deshpande [16] suggests a splitting strategy guided by spatial adjacency. First a graph is constructed with nodes for each sub-region and edges connecting adjacent sub-regions (Fig. 1A). A spanning tree is then randomly sampled from this graph (Fig. 1B), and one or more edges are then randomly deleted to form two or more contiguous sub-regions (Fig. 1C). VCBART utilizes efficient proposal mechanisms based on this strategy to fit a separate ensemble of trees for each spatially varying parameter in model (2), allowing for different spatial clustering patterns for each mixture component. We henceforth refer to the SVC model with parameters estimated using VCBART as SVC BART.

Estimation of the SVC BART model is performed via Markov chain Monte Carlo (MCMC), with each posterior draw yielding different partitions of the study area. Mixture effects might then be summarized for each location using their posterior means and 95% credible intervals (CrIs). BART priors function similarly to *boosting* in that each tree contributes only a small portion to the overall fit, enabling fine tuning of the spatial branching process. Consistent with other BART implementations, regularization priors are used to encourage homogeneity across the entire study area to prevent over-fitting.

Application

In an application of SVC BART, we analyzed 1,468,531 live single-ton births to mothers residing in Georgia. In this sample, the majority of mothers were white (58.1%), and in terms of educational attainment about half (50.7%) reported at least some college experience. Additional demographic information is provided in Table 1.

We fit multiple SVC BART models with the number of trees per ensemble ranging from 5 (simplest) to 50 (most complex). For each model, we kept 2,000 posterior samples after discarding the first 2,000 samples as burn-in. A spatially varying intercept, as well as spatially varying coefficients for quantized versions of $PM_{2.5}$, NO_2 , SO_2 , O_3 , and CO were included. For this analysis, we quantized each exposure into 10 bins, i.e., deciles. Additional covariates modeled using fixed effects included estimated conception date, gestational age, tobacco use, and the parity, age, race, ethnicity, level of educational attainment, and marital status of the mother. We also adjusted for socioeconomic status using Census tract-level estimates of the percentage below the poverty level. The continuous covariates age, tract poverty level, and conception date were modeled using natural cubic splines with 5 degrees of freedom, while gestational age was modeled using indicator variables for the number of weeks.

One of the reasons an SVC model might be appropriate for this analysis is that pollutant concentrations vary across space. We calculated the average (mean) pregnancy-wide concentration of each pollutant within each Georgia county and summarized their distributions in

Table 1
Maternal demographic characteristics.

Characteristic	N	%
Race		
White	853,575	58.1
Black	505,304	34.4
Asian or Pacific Islander	58,706	4.0
American Indian or Alaskan Native	2,460	0.2
Other	48,486	3.3
Ethnicity		
Non-Hispanic	1,252,268	85.3
Hispanic	216,263	14.7
Age		
Less than 25 years	519,590	35.4
25–31 years	568,344	38.7
More than 31 years	380,597	25.9
Education		
Less than 9th grade	73,584	5.0
9th–11th grade	204,226	13.9
12th grade	446,449	30.4
Some college	744,272	50.7

Table 2. Notably, mothers in counties at the 90th percentile of NO_2 and SO_2 were, on average, exposed to more than double the concentration of these pollutants compared to mothers in counties at the 10th percentile. While pollutant concentration typically varies seasonally, some trends are clear, such as CO and NO_2 concentrations being highest in the Atlanta metropolitan area, and NO_2 following the path of I-75 (see Figure S1 of the Supplementary Material).

We compare SVC BART to an ordinary least squares (OLS) model without spatial information, a conditional autoregressive (CAR) model with a spatial random intercept, and a CAR model with spatially varying coefficients in addition to a spatial random intercept. All OLS models were fit using the *rstanarm* R package [17], while CAR models were fit using the integrated nested Laplace approximation (INLA) available via the INLA R package [18]. All models considered include all five pollutants and the aforementioned covariates.

We reference the Widely Applicable Information Criterion (WAIC) for each candidate model as a proxy for leave-one-out cross-validation [19]. The WAIC suggested 25 trees were optimal for SVC BART, so all ensuing comparisons use this specification of the SVC BART model. In Table 3, the effective number of parameters (p_{WAIC}) for SVC BART is nearly twice that of the CAR models, indicating greater model complexity. However, SVC BART achieved the smallest global WAIC, suggesting that this added complexity is justified by the substantial improvement in the model’s ability to describe the observed data. Interestingly, both CAR model specifications fit the data worse than the standard OLS model. These findings support the existence of spatial heterogeneity in the mixture effect while also indicating that the underlying spatial pattern is not adequately captured with a restrictive CAR approach.

The goal of the SVC approach is to improve local mixture effect estimation. With this in mind, we conducted a thorough comparison of

Table 2
Percentiles of county-level mean pregnancy-wide pollutant exposures.

Pollutant	Percentile				
	10th	25th	50th	75th	90th
24-h average PM _{2.5} (µm/m ³)	9.00	9.59	10.22	10.65	10.65
1-h max NO ₂ (ppb)	4.40	5.12	6.60	9.53	9.53
1-h max SO ₂ (ppb)	1.90	2.25	3.14	3.99	3.99
8-h max O ₃ (ppb)	38.98	39.66	40.50	41.22	41.22
1-h max CO (ppb)	0.30	0.31	0.33	0.37	0.37

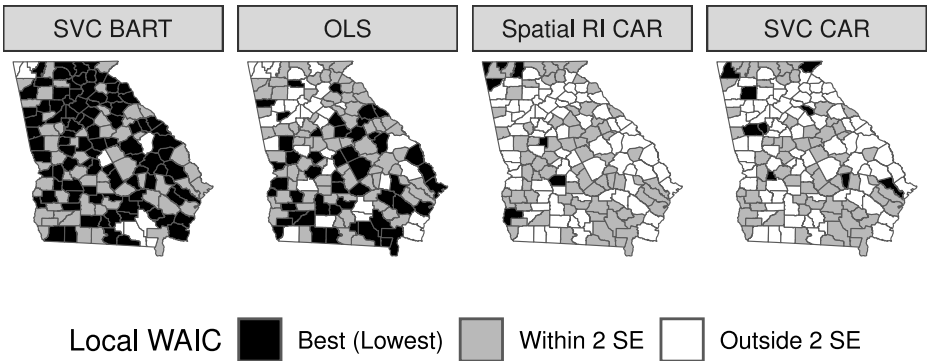


Fig. 2. Comparison of local county-specific WAIC across candidate models. Counties shaded in black had the best (lowest) WAIC under the corresponding model, counties shaded in gray had WAIC within 2 standard errors of the best model, and unshaded counties had WAIC outside 2 standard errors of the best model.

Table 3
Global model comparison.

Model	WAIC	P _{WAIC}	RMSE
SVC BART	21,937,397	411	424.1
OLS	21,939,763	55	424.5
Spatial RI CAR	21,943,089	216	425.8
SVC CAR	21,943,136	212	425.8

WAIC: Widely applicable information criterion.
RMSE: Root mean squared error.
OLS: Non-spatial ordinary least-squares regression.
Spatial RI CAR: Spatial random intercept CAR model.
SVC CAR: Spatially varying coefficient CAR model.

the local WAIC for each individual county. In Fig. 2, we see that SVC BART performs the best in terms of WAIC for 95 out of 159 counties, and has WAIC within 2 standard errors of the best-performing model for 56/64 of the remaining counties. This further supports the use of the SVC BART for estimating local mixture effects.

Estimates of the local mixture effects using the SVC BART model are displayed in Fig. 3. In general, we found stronger negative mixture effects in counties making up the central and northeastern portion of the state. For a one decile increase in all five exposures, the county-specific estimates range from an expected reduction in birth weight of –14.88 grams (95% CrI: –21.24, –9.78) in Whitfield county to an increase of 14.77 grams (95% CrI: 8.73, 20.72) in Coffee county. Of the counties with 95% credible intervals that exclude zero, 21/29 are in a negative direction (see Figure S2 of the Supplementary Material for estimates and credible intervals for these counties).

The common mixture effect for the random intercept CAR model was estimated to be a reduction of –1.81 grams (95% CrI: –2.84, –0.70) per decile increase in all pollutants, and from the OLS model was estimated to be –0.90 grams (95% CrI: –1.59, –0.22). These estimates are slightly attenuated compared to a weighted average of the local mixture effects from the SVC BART model (–1.98, 95% CrI: –3.06, –0.87).

We also compared the estimated local mixture effects against county-level summaries of the exposures and confounders included in the SVC BART model. In general, there was no discernible pattern between the mixture effects and confounders, while point estimates

tended to be higher in counties with lower average CO and NO₂ concentrations (see Figure S3 of the Supplementary Material).

Discussion

We describe varying coefficient models as a useful extension to the popular QGCOMP method for studying environmental exposure mixtures when spatially heterogeneous exposure-response relationships are present. We have shown through an analysis of birth records in Georgia how one might estimate spatially varying parameters in such a model via a Bayesian approach which uses CAR or BART priors.

In addition to the CAR model, other alternatives to SVC BART include geographically weighted regression (GWR) [20] or the SVC model introduced by Gelfand et al. [21]. GWR is a frequentist approach that involves estimating separate weighted least squares regressions at each location, while the SVC model is a Bayesian approach which places Gaussian process (GP) priors on the regression coefficients. Compared to the parametric alternatives, SVC BART is more flexible because it does not require estimating restrictive “nuisance” parameters such as bandwidth (GWR) or spatial correlation (CAR). Although SVC BART can be computationally demanding for large datasets, it avoids the costly inversion of large ($N \times N$) matrices that often renders GP priors impractical. Of these alternatives, the CAR model is the only other adjacency-based approach for estimating spatial heterogeneity (GWR and GP-based SVC models require a prespecified kernel function to estimate proximity between, e.g., region centroids, thus introducing additional complexity).

Deciding whether to model a variable as having a spatially varying effect or to include it as a global confounder should begin with the research question. If spatial variation in a variable’s effect is of substantive interest, it should be modeled as spatially varying. Beyond this, we have some general recommendations. Parameters for variables whose generating process is plausibly spatial (e.g., outdoor environmental exposures) should be allowed to vary across space, whereas individual-level covariates with no clear justification as to why their effect should differ across space (e.g., maternal smoking) can usually be treated as global confounders. For variables lacking spatial or temporal resolution, modeling on the global scale may be necessary for identifiability and stability.

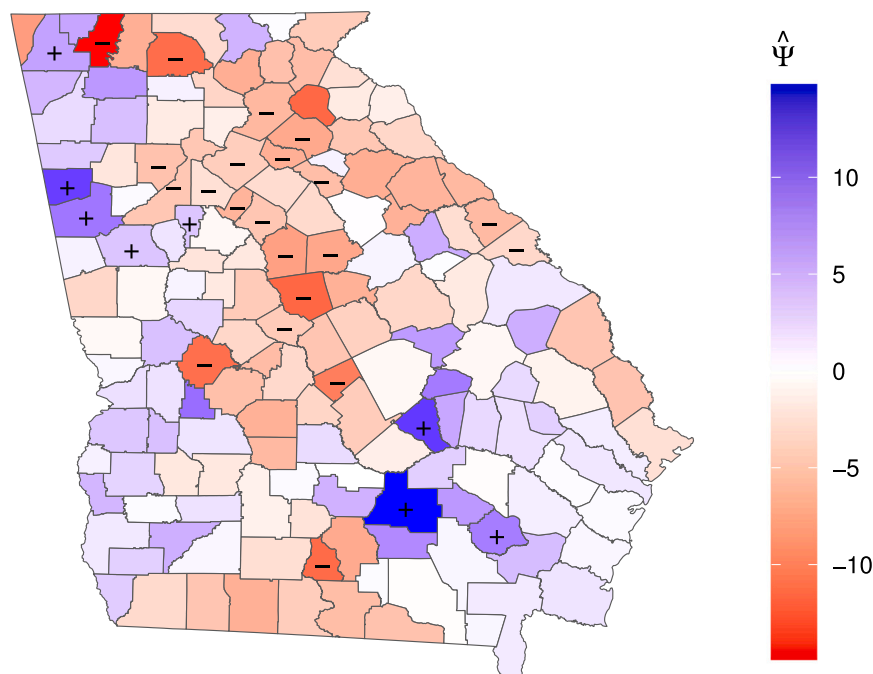


Fig. 3. Local mixture effects estimated using SVC BART. The mean of the posterior distribution of $\Psi(z)$ at each location z is displayed. Counties marked with a “–” have a 95% credible interval entirely below zero, and counties marked with a “+” have a 95% credible interval entirely above zero. See Table S1 of the Supplementary Material for a listing of all county-level mixture effects.

A limitation of our birth weight analysis is the potential for exposure misclassification. This may arise not only from error in the exposure measurements themselves, but also through the mechanism by which we assign pregnancy-averaged pollutant concentrations to each individual. The residential address on file may not be reflective of where the mother spent most her time during the pregnancy, and even when it is, individuals from the same neighborhood may experience different exposures due to unmeasured factors such as occupation, personal lifestyle behaviors, and sources of indoor exposure [22]. However, because pregnant individuals are unlikely to systematically alter their behavior based on outdoor air pollution levels, any misclassification is likely non-differential.

Generally, ambient air pollution contributes little explanatory power for birth weight. The R^2 value for the SVC BART model in the application is 39.89%. Even with a perfect measure of exposure, this value would likely still be low due to the absence of major predictors such as diet and genetics. It is possible that certain pollutants may be more informative if averaged during a critical window of the pregnancy instead of the entire duration. Previous studies of Atlanta, Georgia have identified different associations between air pollution and birth weight for specific months or trimesters of pregnancy [7,8], and several data-driven methods, including some BART-based approaches, have been developed for identifying such susceptible windows [23–26]. On a similar note, spatially varying distributed lag models, such as the one proposed in Warren et al. [27], might be another QGCOMP extension worth exploring.

In this work, we focused on spatial heterogeneity in mixture effects using VCBART’s graph-structured branching process. However, the traditional BART branching process could also be used to model heterogeneity in exposure coefficients as a function of demographic or clinical covariates [28]. For instance, Darrow et al. [7] reported stronger associations between various pollutants and reduced birth weight for Hispanic and non-Hispanic black infants compared to non-Hispanic white infants. The current implementation of the VCBART R package requires the BART ensembles to all use the same set of covariates, but future work could allow covariate selection to differ by exposure.

Conclusions

We extended the QGCOMP framework to account for spatial heterogeneity in environmental mixture effects through the use of VCBART. In our analysis of birth records, we found that elevated concentrations of a mixture of $PM_{2.5}$, NO_2 , SO_2 , O_3 , and CO were associated with reduced birth weight for many Georgia counties from 2005–2016, with the magnitude of the effects varying substantially across space. SVC BART outperformed CAR in capturing this local variation, suggesting that flexible tree-based priors can better represent complex spatial patterns. This approach represents a useful tool for environmental health studies where spatial variation is of interest.

CRedit authorship contribution statement

Jacob Englert: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Howard Chang: Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Howard H Chang reports financial support was provided by National Institute of Environmental Health Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Institute of Environmental Health Sciences, United States (NIEHS) of the National Institutes of Health (NIH) [grant number R01ES028346, R01ES037289].

Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.annepidem.2025.10.003>.

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